

Day 11: DAX Fundamentals & Calculated Columns vs Measures

📌 **Definition to Remember**

DAX (Data Analysis Expressions) is the native formula and functional expression language of Power BI used to create custom calculations, dynamic business metrics, and tabular evaluations.

1. Calculated Columns vs Explicit Measures

Feature	Calculated Column	Explicit DAX Measure
Evaluation Timing	Calculated during data refresh / model load.	Evaluated on-the-fly dynamically when visuals render.
Storage (RAM)	Stored in memory and increases file size.	Consumes ZERO storage ; calculated dynamically in CPU.
Context	Evaluates in Row Context (row-by-row).	Evaluates in Filter Context (respects slicers & visual filters).
Best Used For	Slicer fields, row categorizations, age buckets.	All business KPIs, numeric metrics, ratios, & totals.

2. DAX Syntax & Operators

```
// Measure Syntax: MeasureName = FUNCTION(Table[Column])
Total Sales = SUM(Fact_Sales[SalesAmount])

Total Quantity = SUM(Fact_Sales[Quantity])
```

- Arithmetic:** +, -, *, /
- Comparison:** =, <=>, <, >, <=, >=
- Logical:** && (AND), || (OR), IN (Set membership), NOT

3. Golden Rule of Power BI Development

Always use Explicit DAX Measures for aggregations and numeric metrics. Avoid implicit auto-sums (Σ Column) and avoid unnecessary calculated columns to keep model RAM lean.

★ Must-Master Skills & Interview Points

1. Clear distinction between Calculated Columns (static, row context, RAM heavy) and Measures (dynamic, filter context, CPU efficient).
2. Why explicit measures enable dynamic drilldowns and interactive visual recalculation.
3. Standard naming conventions for clean DAX code.

⚡ Quick Recall

Calculated Column = Stored in RAM, Row Context | DAX Measure = Computed on-the-fly, Filter Context, Gold Standard